

Quiz: Chapter 2

Name(s): _____

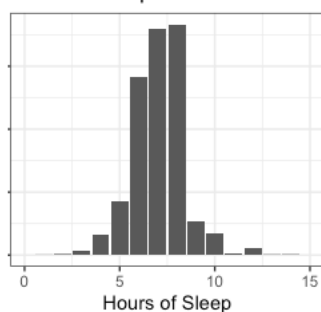
Quiz Information

Complete all of the questions and show your work when necessary. I encourage you to work with a partner on this quiz, but do not refer to your notes. Only submit one quiz per group.

Scenario: Amount of Sleep

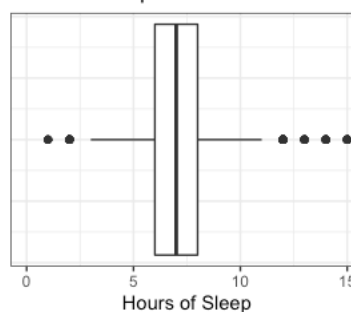
The National Health Interview Survey (NHIS) is a survey collecting information on the health, health care access, and health behaviors of United States citizens. One question on the 2018 survey asked the respondents how many hours of sleep they usually get per night. The summary statistics and plots inspired by this study are below.

Histogram for Number of Hours Slept



Source: NHIS

Boxplot for Number of Hours Slept



Source: NHIS

Mean	Standard Deviation	Number of Respondents
7.3	1.2	247

Researchers are interested in determining if the amount of sleep that Americans are getting is different than the typical 8 hours.

- 1) (2pts) Describe the distribution for the number of hours of sleep. Can we assume normality?

2) (3 pts) Write down the null and alternative hypothesis, in symbols, that reflects the research question.

3) (5 pts) Calculate the standardized statistic and show your work. Recall that the formula for the standardized statistic is given by

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

4) (2 pts) Based on your standardized statistic, what conclusion would you tell the researchers?

5) (1 pts) Is it possible that this study made a Type I or Type II error? (*Yes / No*)

6) (1 pts) If so, which type of error would this scenario make? (*Type I / Type 2*)

7) (2 pts) Briefly state how this type of error would affect your conclusion in terms of the context of the scenario.